Fake News Detection Using AI

GenAI Internship — PBEL Virtual Internship

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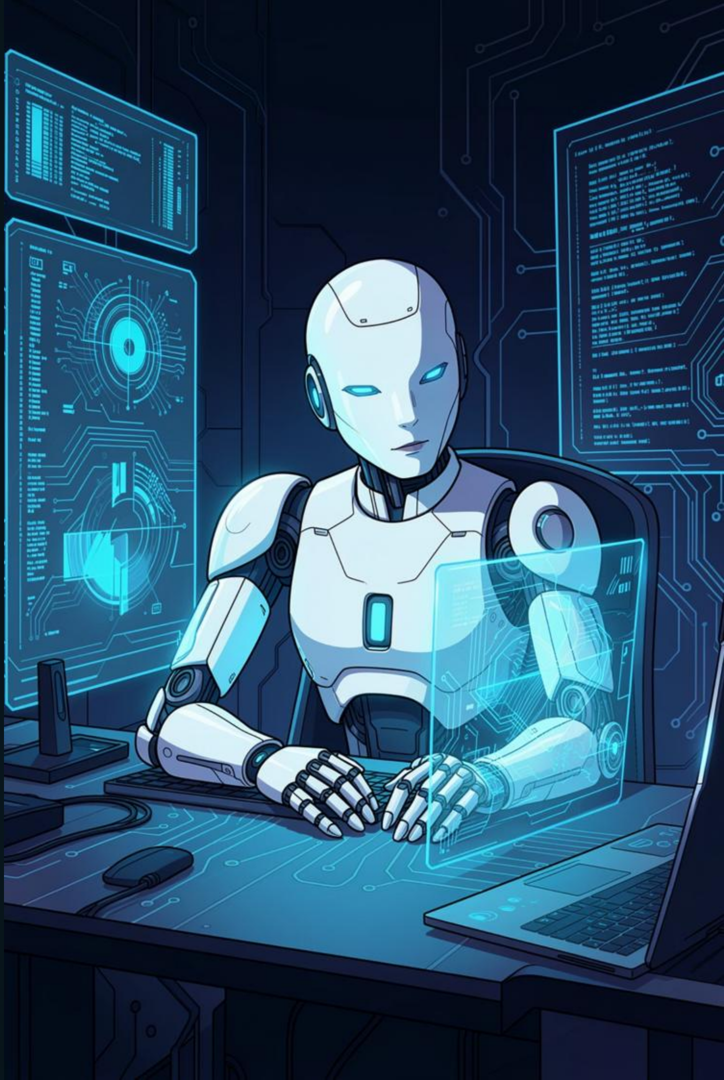


The Challenge of Fake News

Fake news spreads rapidly across social media, influencing public opinion at scale. Manual verification is increasingly impossible as content volume grows exponentially.

The Problem

Volume of digital content makes manual fact-checking impractical

The Solution

AI automates detection by analyzing patterns and linguistic features

The Goal

Distinguish real from fake news with remarkable accuracy

Problem Statement

Detection Challenge

Accurately classify news articles as real or fake using automated systems

AI Generation

Generate synthetic fake news headlines to test detection capabilities

Real-Time System

Provide instant predictions as news emerges in the digital landscape

Project Objectives

1. **Build Detection System**
   Develop a comprehensive fake news detection framework using machine learning

2. **Apply NLP Techniques**
   Implement Natural Language Processing to analyze text patterns and features

3. **ML Classification**
   Use machine learning algorithms to classify news articles accurately

4. **Generative AI**
   Implement Generative AI to create fake news examples for testing

5. **User Interface**
   Create an intuitive, user-friendly interface for easy interaction

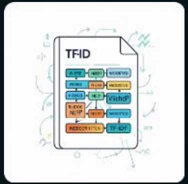
Technologies Used

Python

Core programming language for implementation and integration

Streamlit

Framework for building the interactive user interface



NLP (TF-IDF)

Term Frequency-Inverse Document Frequency for text vectorization

News Dataset

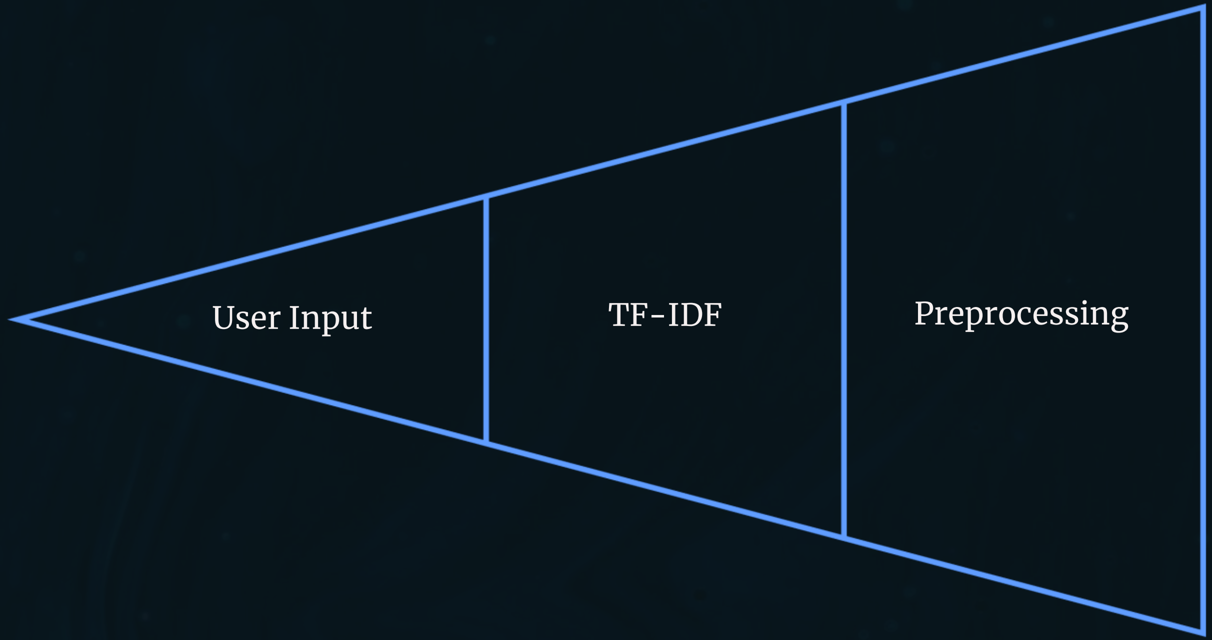
Comprehensive collection of verified fake and real news articles

Machine Learning

Naive Bayes classifier for efficient text classification

System Architecture

The detection system follows a sequential pipeline that processes news articles through multiple stages to deliver accurate predictions.



Each stage transforms raw input into structured features, enabling the Naive Bayes classifier to deliver real-time, accurate fake news predictions.

Methodology

01

Data Collection

Gather comprehensive dataset of fake and real news articles from verified sources

02

Preprocessing

Remove noise, normalize text, handle missing values, and prepare data for analysis

03

Feature Extraction

Convert text to numerical features using TF-IDF vectorization techniques

04

Model Training

Train Naive Bayes classifier and evaluate performance using test data

05

Deployment

Implement Streamlit interface for real-time predictions and user interaction

Model & Results

Naive Bayes Classifier

Selected for its speed and efficiency with text data. This probabilistic classifier works exceptionally well for document classification tasks.

- Fast training and prediction times
- Effective with high-dimensional data
- Requires minimal parameter tuning
- Performs well with limited training data

90%

Accuracy

Achieved in classifying news articles as real or fake

~0s

Real-Time

Instant results delivered via the Streamlit interface

Generative AI & Applications

The system includes a **Generative AI module** that creates synthetic fake news headlines — demonstrating Generative AI while providing test cases for the detection system.

Social Media Monitoring

Track and verify news spreading across platforms like Twitter, Facebook, and Instagram

News Verification

Help journalists and fact-checkers validate sources before publication

Cybersecurity

Detect malicious disinformation campaigns and propaganda operations

Conclusion & Future Scope

Conclusion

NLP combined with Machine Learning effectively detects fake news with **~90% accuracy**. The system provides real-time results through an intuitive interface. The integration of Generative AI demonstrates both the problem and solution — a comprehensive tool for combating misinformation.

Future Enhancements

- **BERT & Deep Learning** — Implement transformer-based models for improved understanding
- **Higher Accuracy** — Refine models and expand training datasets
- **Multilingual Support** — Add capability to detect fake news in multiple languages

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Thank You

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